

Project FORESIGHT

Demand Forecasting & Inventory Risk Intelligence Platform (NorthBay Living)

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github.com/arvijadhav1209-hub/foresight foresight-gctc.onrender.com		

1. Executive Summary

In retail and omni-channel commerce, inventory mismanagement drives significant financial drag through two major channels: **stockouts** (lost immediate revenue and customer churn) and **overstocking** (tied-up working capital, elevated holding costs, and margin degradation via forced clearance markdowns).

Project FORESIGHT is an end-to-end predictive analytics and decision-support system built for NorthBay Living across 12 high-velocity everyday home product SKUs over a 2-year transactional horizon. The system delivers an automated ETL pipeline, a 6-week rolling time-series forecast model using [HistGradientBoostingRegressor](#) (**14.5% WAPE**, a **19.7% accuracy boost** over seasonal-naïve benchmarks), a risk-scoring matrix at a 90% service level, and an interactive executive planning dashboard deployed on Render.

Key Business Impact: Flagged **₹7.28 Lakhs in sales-at-risk** across 5 stockout-prone products and identified **₹2.14 Lakhs in surplus working capital** trapped in overstocked goods, providing actionable replenishment quantities directly to operations.

2. Problem Statement & Objectives

NorthBay Living manages diverse catalog lines spanning Kitchenware, Bedding, Home Decor, and Small Appliances with supplier lead times between 7 and 30 days. Operations teams lacked unified tools to project demand across marketing events or quantify the financial stakes of reordering too late vs. buying too much.

Core Deliverables:

- **D1 (Data Pipeline & Hygiene):** Ingest and clean 4 transactional datasets (duplicates, missing prices, string anomalies) and create an indexed SQLite database.
- **D3 (Machine Learning Forecast):** Predict SKU-level demand for the next 6 weeks using strictly leak-free historical, calendar, and promotional features.
- **D4 (Risk Scoring & Decisions):** Categorize inventory into a 4-quadrant decision matrix (Reorder Now, Markdown / Clear, Watch, Healthy) with monetary risk estimation.
- **D5 (Web Dashboard & Security):** Develop an interactive, role-based planning tool with PBKDF2 cryptographic password hashing and deploy to cloud production.

3. Machine Learning Demand Forecasting

The forecasting engine predicts weekly sales for $H = 6$ weeks ahead. To avoid lookahead bias, all features use only information available up to the forecast origin date:

- `last1` : Immediate prior week demand.
- `m4, m8` : 4-week and 8-week moving averages of demand velocity.
- `ly, ly3` : Demand from identical week last year (t-52) and centered 3-week window.
- `promo, hol` : Future known promotional events and official public holiday counts.
- `woy` : ISO calendar week of the year to capture annual seasonality.

Backtest Performance (6 Rolling Rounds × 6 Weeks)

Evaluation Metric	Seasonal-Naive Baseline	Gradient Boosting Model	Relative Delta
WAPE (Weighted Absolute % Error)	18.00%	14.46%	+19.70% Improvement
Forecast Bias	—	-0.015	Unbiased (no systematic over/under forecasting)
Production Selection Rule	Discarded	Active Model	Model verified superior to baseline

4. Inventory Risk Scoring & Decision Matrix

Safety stock is dynamically sized for a **90% Cycle Service Level** ($Z = 1.28$) considering weekly demand volatility (σ_{13}) and supplier lead times:

$$\text{Safety Stock} = 1.28 \times \sigma_{13} \times \sqrt{\text{Lead Time Weeks}}$$

Total replenishment need is compared against supply (On Hand + On Order) to compute the **Cover Ratio** and **Weeks of Supply**, populating the 4-box decision grid:

Product Name	Category	Weeks of Stock	Cover Ratio	Decision Quadrant	Sales at Risk (₹)	Locked Capital (₹)	Suggested Order
Steel Water Bottle	Kitchen	0.9	0.64	Reorder now	₹18,772	₹0	251 units
Non-stick Pan	Kitchen	1.0	0.42	Reorder now	₹98,713	₹0	179 units
Woolen Blanket	Bedding	1.0	0.31	Reorder now	₹2,39,878	₹0	134 units
Electric Kettle	Small Appliances	2.1	0.43	Reorder now	₹2,19,916	₹0	185 units
Room Fan	Small Appliances	4.2	0.76	Reorder now	₹1,10,100	₹0	85 units
Mixer Grinder	Small Appliances	6.5	0.95	Watch / volatile	₹40,186	₹37,800	47 units
Cotton Bedsheet	Bedding	7.9	3.40	Markdown / clear	₹0	₹1,20,000	0 units
Wall Clock	Home Decor	14.1	6.00	Markdown / clear	₹0	₹56,256	0 units
Ceramic Coffee Mug	Kitchen	3.0	1.93	Healthy	₹0	₹0	0 units
Soft Pillow Set	Bedding	3.9	1.74	Healthy	₹0	₹0	0 units
Table Lamp	Home Decor	4.3	1.85	Healthy	₹0	₹0	0 units
Scented Candle	Home Decor	2.8	1.77	Healthy	₹0	₹0	0 units

Product Name	Category	Weeks of Stock	Cover Ratio	Decision Quadrant	Sales at Risk (₹)	Locked Capital (₹)	Suggested Order
TOTAL / SUMMARY	12 Products	—	—	5 Reorder, 2 Clear	₹7,27,565	₹2,14,056	882 units

5. Deployment & System Architecture

- **Interactive Web Dashboard:** Developed with Streamlit and Plotly, delivering responsive data visualization and self-healing SQLite database instantiation.
- **Enterprise Security:** Implemented PBKDF2 HMAC-SHA256 password hashing with 100,000 rounds and unique per-user salts. Roles: `admin` (full configuration) and `ops` (planning).
- **Cloud Infrastructure:** Deployed as a web service on Render (`foresight-gctc.onrender.com`) using `render.yaml` and `Procfile` .
- **Version Control:** Managed via Git and published on GitHub at `arvijadhav1209-hub/foresight` .

6. Conclusion

Project FORESIGHT delivers an automated, mathematically rigorous, and commercially actionable decision-support solution for NorthBay Living. By shifting from reactive replenishment to predictive inventory risk management, operations teams safeguard critical revenue, prevent customer attrition, and liberate stagnant working capital.

Submitted in partial fulfillment of the Internship Requirements at **Zidio Development**.
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